

COMICCRAFT

DOCUMENT 13: SCALABILITY & FUTURE PLAN

Current System Limitations

S.NO	LIMITATION	IMPACT	PRIORITY TO ADDRESS
1	Image generation is GPU-intensive and runs synchronously	Slower on CPU-only machines; blocks the request thread; limits concurrent users	High
2	No user authentication or comic history saving	Users cannot save, revisit, or share their generated comics. No personal gallery	Medium
3	Single AI model provider (Gemini only)	Vendor lock-in risk; no fallback if Gemini API changes pricing/availability	Low
4	File-based storage with no database	No search, filtering, categorization, or multi-user data management capabilities	Medium

Scalability Plan

S.NO	SCALABILITY ASPECT	CURRENT STATE	PROPOSED UPGRADE / SOLUTION
1	User Load	Single-user local development; synchronous request handling	Deploy on cloud platform (Render, Railway, Hugging Face Spaces) with auto-scaling. Add async task queue
2	Data Storage	File-based (static/panels, static/exports) - temporary, no persistence	Add cloud storage (AWS S3 / Cloudinary) for images. PostgreSQL/MongoDB for user comic history and metadata
3	Performance	Synchronous pipeline - blocks until all 5 images are generated	Add async task queue (Celery + Redis) for image generation. Implement caching for repeated prompts. Stream partial results
4	Security	No authentication; API keys in local .env	Add JWT-based authentication, rate limiting (100 req/hour/user), API key rotation, HTTPS enforcement

Future Roadmap

PHASE	PLANNED FEATURE / ENHANCEMENT	TARGET TIMELINE	EXPECTED IMPACT
Phase 2	User authentication (JWT), comic history saving, personal gallery	Q2 2027	Users can create accounts, save comics, revisit past creations, share with unique URLs

Phase 3	Multi-language support (prompts and output in 10+ languages)	Q3 2027	Global accessibility - users worldwide can create comics in their native language
Phase 4	Image-to-comic feature (upload photo → convert to comic style)	Q4 2027	New creative use case attracting photography enthusiasts and cosplay communities
Phase 5	Social sharing, community gallery, comic remixing/collaboration	Q1 2028	Viral growth through user-generated content sharing, community engagement